$P_1V_1 = P_2V_2$ (1) $P_{total} = P_1 + P_2 + P_3$ (2) $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ (3) $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ (4) $PV = nRT$ (5) $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ (6) $\frac{V_1}{n_1} = \frac{V_2}{n_2}$ (7)

12

Chemistry 3A – Fall 2025 Tuesday Version: **Section 25810**

Name Skyelee ID 1073585

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

b $\frac{V_1}{T_1} = \frac{V_2}{T_2}$	**(a)** Boyle's Law **(b)** Charles' Law **(c)** Gay-Lussac's Law **(d)** Avogadro's Law **(e)** Ideal Gas Law
c $PV = nRT$	
e "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."	
a $P_1V_1 = P_2V_2$	

2

2. Standard Temperature and Pressure (STP) (3 pts) 760 mmHg or Torr & 273.15 K

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

The numerical value and unit for standard pressure used in the course is 760 mmHg.

1.5

(b) What is standard temperature in Celsius (°C)? (1 pt)

The standard temperature in Celsius is 1.8°C.

(c) Convert that standard temperature into Kelvin (K). (1 pt)

1.8 + 273.15 = 274.95 K

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

2.5

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $V_1 = 3.50\,L$ $T_1 = 25°C$ $V_2 = 4.25\,L$ $T_2 = ?$

$T_2 = \frac{T_1 V_2}{V_1}$

$\frac{25°C \times 4.25\,L}{3.50\,L}$

$= 30.357142\overline{86}$

$- 273.15$

$= -242.79$

$= -240°C$

4. Kinetic Molecular Theory (3 pts)

According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

4 The volume and atmospheric pressure collides and its molecules collides with the temperature. There is an increase in pressure and energy of gas particles as there is a collision between them and the container walls.

5. Ideal Gas Law Calculation (5 pts)

Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$PV = nRT$ $P = \frac{nRT}{V}$ $\frac{0.150 \text{ mol } CO_2 \times 0.08205 \text{ L} \times \text{atm} / (\text{mol} \times \text{K}) \times 310 \text{ K}}{2.50 \text{ L}}$

4 $P = ?$

$V = 2.50 \text{ L}$

$n = 0.150 \text{ mol } CO_2$

$R = 0.08205 \frac{L \times atm}{mol \times K}$

$T = 310 \text{ K}$

$= 1.52613 \text{ K atm } CO_2$

$= \boxed{1.5 \text{ K atm } CO_2}$

$P_{\text{total}} = \frac{n_1 + n_2}{P}$

$P_{\text{total}} = \frac{n_1 + n_2}{P_1 + P_2}$? $n_1 + n_2 = \frac{n_3}{P_1}$

6. Partial Pressures and Mole Fraction (8 pts)

A gaseous mixture in a sealed container consists of **0.400** moles of nitrogen gas (N_2) and **0.100** moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be **1.25** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

$0.400 + 0.100 = 0.500$ moles

The total number of moles is 0.500 moles of gas.

1

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

$\frac{1.25 \text{ atm}}{0.100 \text{ mol}_{O_2}} = \boxed{12.5 \text{ atm mol } O_2}$

$\frac{1.25}{0.100 + 0.400} = 0.892857143$

$= 0.893 \text{ atm mol}_{O_2}$?

$\frac{P}{n}$? $\frac{P}{n_1 + n_2}$

(c) Calculate the partial pressure (P_{O_2}) of oxygen gas in the mixture. *(3 pts)*

$P_{O_2} + P_{N_2} = P_{\text{total}}$ $0.500 P_{\text{total}} - 0.400 P_{N_2} = \boxed{0.100 P_{O_2}}$?

$P_{\text{total}} - P_{N_2} = P_{O_2}$?

$\frac{n}{P}$? $\frac{n_1}{P_1} = \frac{n_2}{P_2}$

1.25 atm ↑ 0.500 mol gas

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

P_n

If the total pressure increases, the mole fraction of O_2 will have a collision with the mole fraction of N_2. This is because it forces the two gases to interact with one and another.

Pressure (P) Volume (V) Temp (T) n (mol) $R = 0.082058$

$P_1V_1 = P_2V_2$ $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ $PV = nRT$ $\frac{V_1}{n_1} = \frac{V_2}{n_2}$ $P_{Total} = P_1 + P_2 + P_3 \ldots$

$d = \frac{M}{V}$

Chemistry 3A – Fall 2025 24.5 Tuesday Version: **Section 25810**

Name Shane Xiong ID 1012185

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

b $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
e $PV = nRT$
d "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
a $P_1V_1 = P_2V_2$

4

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 atmosphere (atm) = 760 Torr, 101325 Pa 1 Torr = 1 mm Hg, 1.3332×10^2 Pa

(b) What is standard temperature in Celsius (°C)? (1 pt)

0°C

3

(c) Convert that standard temperature into Kelvin (K). (1 pt)

$0°C + 273.15 = \boxed{273.15\ K}$

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

5

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$ $V_1 = 3.50\ L$ $T_1 = 25°C$ $V_2 = 4.25\ L$ $T_2 = ?$

$= 25°C + 273.15$

$= 298.15\ K$

↑ yes, because all units must be metric system

2. $\frac{T_1}{V_1} = \frac{T_2}{V_2} \cdot V_2$

$T_2 = \frac{T_1V_2}{V_1} = \frac{(298.15\ K)(4.25\ L)}{3.50\ L} = 362.04\ K - 273.15 = \boxed{88.9°C}$

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

1 The energy increases, increasing pressure within the container, from increasing gas particle speed, until ultimately exploding. (volcano)

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$PV = nRT$ $n = 0.150\,mol$ $V = 2.50\,L$ $T = 310\,K$ $R = 0.08205$ $P = ?$

$$P = \frac{nRT}{V} = \frac{(0.150\,mol)(0.08205\,\frac{L \cdot atm}{mol \cdot K})(310\,K)}{2.50\,L} = \boxed{1.53\,atm}$$

5

$\frac{V}{n} = P$ $n = \frac{V}{P}$ $V = P \cdot n$

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of **0.400** moles of nitrogen gas (N_2) and **0.100** moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be **1.25** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

$n_{Tot} = n_1 + n_2 = 0.400\,mol\,N_2 + 0.100\,mol\,O_2 = \boxed{0.500\,mol}$

1

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

$\frac{P}{n} = 0.5$ $\frac{mol\,O_2}{n_{tot}} \times P_{tot}$

(c) Calculate the partial pressure (P_{O_2}) of oxygen gas in the mixture. *(3 pts)*

2.5 $\frac{0.100\,mol\,O_2}{0.500\,mol} \times 1.25\,atm = \boxed{0.25\,atm}$

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

1.5 The partial pressure of the mole fraction of O_2 will stay constant because there will remain the same amount of gas in the container.

Chemistry 3A – Fall 2025 13 Tuesday Version: **Section 25810**

Name Marcella Perez ID 1096135

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

d $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
e $PV = nRT$
c "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
b $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

1

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 atm, 0°C 1 mol = 22.4 L

3

(b) What is standard temperature in Celsius (°C)? (1 pt)

0 °C

(c) Convert that standard temperature into Kelvin (K). (1 pt)

K = °C + 273.15 K = 273.15

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?) ← No

$P_1V_1 = P_2V_2$

2.5

$25°C \cdot 3.50L = P_2 \cdot 4.25L$ $P_2 = 20.588235294°C$

$P_2 = 21 °C$

$\frac{25 °C \cdot 3.50}{4.25} = P_2$

4. Kinetic Molecular Theory (3 pts)

According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

0.5 Kinetic energy builds up, allowing for particles to heat up (idk)

5. Ideal Gas Law Calculation (5 pts)

Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

$PV=nRT$

$P(2.50L) = .150\,\text{mol}\,CO_2 \cdot \frac{.08205\,L\cdot atm}{mol\cdot K} \cdot 310\,K$

5 $P=$

$V=2.50L$

$n=.150\,\text{mol}\,CO_2$

$R=.08205\,\frac{L\cdot atm}{mol\cdot K}$

$T=310K$

$P(2.50L) = 3.815325\,CO_2\,L\cdot atm$

$P=1.52613\,CO_2\,atm$

$P=1.53\,CO_2\,atm$

6. Partial Pressures and Mole Fraction (8 pts)

A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles ($\boldsymbol{n}_{\text{tot}}$) of gas in the container? *(1 pt)*

$n_{tot} = \frac{x}{P}$ ~(0.)~ *literally just studied this too and I already forgot how to*

$(.400+.100)(1.25) = .625\,\text{mol}$

1 (b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

$\frac{.100}{.625} = .16\,\text{mol}$

(c) Calculate the partial pressure ($\boldsymbol{P}_{O_2}$) of oxygen gas in the mixture. *(3 pts)*

$\frac{.16\,\text{mol}}{1.25\,\text{atm}} = .128\,\frac{\text{mol}}{\text{atm}}$

2.5

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

The gas will decrease. More pressure = less volume

Chemistry 3A – Fall 2025 7 Tuesday Version: **Section 25810**

Name Delona Adams ID 1111240

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

a $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
d $PV = nRT$
e "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
b $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

760 Torr, 101325 pa

(b) What is standard temperature in Celsius (°C)? (1 pt)

-32

(c) Convert that standard temperature into Kelvin (K). (1 pt)

244.64827 (K)

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

0.5

P 3.50L = n 4.25L 3.50L

$\frac{14.875\,L}{3.50} \rightarrow$ 4.25

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

3

They start to either cancel eachother out or react in the container walls, like bouncing off of eachother

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

1

$R = 0.08205 \frac{2.50L}{0.150 \cdot 310K}$

64182 LK

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

0.5

500

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

$1.25 atm = \frac{0.100}{0.400}$ 0.125

(c) Calculate the partial pressure (P_{O_2}) of oxygen gas in the mixture. *(3 pts)*

2

0.25

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

the mole fraction will keep increasing

8.5

Chemistry 3A – Fall 2025 Tuesday Version**: Section 25810**

Name Epifania Cirilo-Lopez ID 090,1937

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

b $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
d $PV = nRT$
c "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
a $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

0.082058

(b) What is standard temperature in Celsius (°C)? (1 pt)

= 32°C

1

(c) Convert that standard temperature into Kelvin (K). (1 pt)

305.15 K

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

$\frac{3.50_{V_1}}{298.15T_1} = \frac{4.25_{V_2}}{}$

25°C + 273.15 = 298.15

3

$V_1 = 3.50$

$V_2 = 4.25$

$T_1 = 298.15$

$T_2 = ?$

$\frac{(4.25)(298.15)}{3.50} = 362.04$

362.04 − 32 / 1.8 =

C° = 183.36

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

1.5 They are able to slide past eachother and fit to the shape of the container.

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

0.08205

$$= \frac{2.50_L \cdot 28,01}{0.150 \text{ moles} \cdot 310} = \frac{70.025}{46.5} = 18.35$$

1

R = 18.35 ?

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

$n_{tot} = 0.50$

1

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

(c) Calculate the partial pressure (P_{O_2}) of oxygen gas in the mixture. *(3 pts)*

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

Chemistry 3A – Fall 2025 7.5 Tuesday Version: **Section 25810**

Name Rosa M Contreras ID 0195236

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

b $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
d $PV = nRT$
C "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
a $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 atm → 760 torr → 101325 Pa

2.5

(b) What is standard temperature in Celsius (°C)? (1 pt)

(°F - 32) / 1.8

(c) Convert that standard temperature into Kelvin (K). (1 pt)

°C + 273.15

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

Given $V_1 = 3.50\,L$, $T_1 = 25°C$, $V_2 = 4.25\,L$ ~~Convert C→K 25 + 273 = 298~~

Find T_2

Plan $T_2 = \frac{V_1 \times T_1}{V_2}$ $T_2 = \frac{3.50\,L \times 25^{\circ}C}{4.25\,L} = 20.588235\overline{29}$

$T_2 = 20.6^{\circ}C$

2

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of **0.400** moles of nitrogen gas (N_2) and **0.100** moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be **1.25** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

(c) Calculate the partial pressure (P_{O_2}) of oxygen gas in the mixture. *(3 pts)*

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

8.5

Chemistry 3A – Fall 2025 Tuesday Version**: Section 25810**

Name Sabrina Garcia ID 0490011

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

b $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
e $PV = nRT$
d "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
C $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

3

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

(b) What is standard temperature in Celsius (°C)? (1 pt)

The standard temperture in Celsius is 1.8 °C

(c) Convert that standard temperature into Kelvin (K). (1 pt)

1.8°C + 273.15 = 274.95 °K

1

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

1

$\frac{(3.50L)(4.25L)}{25°C} = 0.595\ °C$

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.
They will gain more electrons and increase in size

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

3.5

12.011 + 15.999

$\frac{2.50\text{ L} \times 28.01}{0.150 \times 310} = \frac{70.025}{46.5} =$ 1.50 atm

1.53

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles ($\boldsymbol{n}_{\text{tot}}$) of gas in the container? *(1 pt)*

$\frac{0.100}{0400} =$ 0.25 moles ✓

(b) Determine the mole fraction ($\boldsymbol{\chi}$) of oxygen gas (O_2). *(2 pts)*

$\frac{0.100 \text{ moles}}{6.0221 \times 10^{23}} = 1.66$

(c) Calculate the partial pressure ($\boldsymbol{P}_{O_2}$) of oxygen gas in the mixture. *(3 pts)*

$\frac{0.100}{2} =$ 0.05

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*
The mole fraction of O_2 will decrease because the pressure of the gas has increased.

15

Chemistry 3A – Fall 2025 Tuesday Version: **Section 25810**

Name Jose Sanchez ID 10354163

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

4

B $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
e $PV = nRT$
d "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
A $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 atm

2

(b) What is standard temperature in Celsius (°C)? (1 pt)

0

(c) Convert that standard temperature into Kelvin (K). (1 pt)

°C + (K) = K

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

1

$3.50 \times 25 = 87.5°C$ $\frac{(3.50)(25°C)}{4.25} = 21°C$

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

2.5 There is No energy being LOST if temperature is being held constant

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

2.5 (0.150)(2.50)(310)(.08205

9.5383 atm

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles ($\boldsymbol{n}_{tot}$) of gas in the container? *(1 pt)*

1 $P_1.400 + P_2.100 = P_t.500$

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

2 $Pt.5 / P_2.1 = 0.202$

(c) Calculate the partial pressure ($\boldsymbol{P}_{O_2}$) of oxygen gas in the mixture. *(3 pts)*

$$\frac{0.100 \times 1.25}{0.2} = .1625$$

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

Since the pressure is being increased the volume will be lower.

3

Chemistry 3A – Fall 2025 Tuesday Version**: Section 25810**

Name Vanessa Arredondo ID 06185559

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

B $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
E $PV = nRT$
C "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
A $P_1V_1 = P_2V_2$

3

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

anions

(b) What is standard temperature in Celsius (°C)? (1 pt)

(c) Convert that standard temperature into Kelvin (K). (1 pt)

39.10

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

3.50 L

35°C

4.25 L

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

It evaporates and dissolves

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

$0.150 \frac{2.50}{310\text{K}}$ 9.8287

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

0.04 mol

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

0.200 mL

(c) Calculate the partial pressure (P_{O_2}) of oxygen gas in the mixture. *(3 pts)*

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

23

Chemistry 3A – Fall 2025 Tuesday Version: **Section 25810**

Name Oluwapelumi Tay ID _____

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

(B) $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
(E) $PV = nRT$
(C) "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
(A) $P_1V_1 = P_2V_2$

3

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

STP = 22.4L; Numerical value is 22.4 and unit is volume (L)

(b) What is standard temperature in Celsius (°C)? (1 pt)

Standard temperature in Celsius is 0°C = 273K

2

(c) Convert that standard temperature into Kelvin (K). (1 pt)

0°C = 273K

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

$V_1 = 3.50L$

$T_1 = 25°C + 273K = 298K$

$V_2 = 4.25L$

$T_2 = ?$

2

3

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} = \frac{3.50L}{(298K)(4.25L)} = 0.00276$$

$$= 0.003 \text{ or } 3.0 \times 10^{-3} K$$

$$T_2 = \frac{V_1}{T_1 V_2}$$

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

The energy of gas particles elastically collide with each other and the container walls. The pressure inside the contained gas = the pressure outside the container so it stays the same.

3

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$PV = nRT \quad P = ?$

$$P = \frac{0.150 \times 0.08205 \times 310}{2.50}$$

$$P = 1.53 \text{ atm}$$

5

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

$n_{tot} = 0.400 + 0.100$

$= 0.500$ moles

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

Mole fraction of O $= \frac{0.100}{0.500} = 0.200$ mol

7

(c) Calculate the partial pressure (P_{O_2}) of oxygen gas in the mixture. *(3 pts)*

Partial Pressure $= P_A = P_{Total} X_A$

$= 1.25 \times 0.200$

$= 0.250$ atm

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

The mole fraction of O_2 decreases due to the force exerted to release pressure (Kinetic theory) As the pressure increases, mole fraction decreases because the total mole fraction will always equal 1.

Chemistry 3A – Fall 2025 5.5 Tuesday Version: **Section 25810**

Name Emily Mousesian ID 0822239

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

e $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
b $PV = nRT$
d "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
A $P_1V_1 = P_2V_2$

2

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 atm, 760 Torr, 101325 Pa

(b) What is standard temperature in Celsius (°C)? (1 pt)

2 °32

(c) Convert that standard temperature into Kelvin (K). (1 pt)

32 + 273.15 = 305.15 K

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

(3.50 L)(4.25L) = 14.875°C

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

As the Gas Temp goes up, the energy increases while the volume decreases. AS well as if it decreases the volume increases.

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$R = (0.08205 \cdot \frac{2.50 \cdot 1\,atm}{0.150 \cdot 310} = 423.925$ Atm

1.5

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles ($\boldsymbol{n}_{tot}$) of gas in the container? *(1 pt)*

0.400

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

$O = 15.999 \times 2 = 31.998 \rightarrow 32$

0.100

$\frac{32}{0.100} = 320$

$\frac{15999}{500}$

(c) Calculate the partial pressure ($\boldsymbol{P}_{O_2}$) of oxygen gas in the mixture. *(3 pts)*

0.400 Mol

0.100 Atm 1.25

0.100 Mole = 1 Mole is 6.0221×10^{-34}

$\frac{0.100}{1\,Atm}$ 1 Atm

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

The Mole fraction of O_2 will decrease

22

Chemistry 3A – Fall 2025 Tuesday Version: **Section 25810**

Name JUSTIN VINK ID 1070339

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

b $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
e $PV = nRT$
c "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
a $P_1V_1 = P_2V_2$

~~(a) Boyle's Law~~
~~(b) Charles' Law~~
~~(c) Gay-Lussac's Law~~
~~(d) Avogadro's Law~~
~~(e) Ideal Gas Law~~

3

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 atm

3

(b) What is standard temperature in Celsius (°C)? (1 pt)

0 °C

(c) Convert that standard temperature into Kelvin (K). (1 pt)

273.15 K

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

3.50L 25°C **30.4°C**

2

4.25L _°C

$\frac{106.25}{3.50} = \frac{3.50x}{3.50}$

$\frac{3.50L}{25°C} = \frac{4.25L}{x°C}$

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

1

THE ENERGY OF GAS PARTICLES INCREASES THE MORE THEY COLLIDE WITH EACH OTHER.

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

5

$PV = nRT$

$P(2.50) = (0.150)(0.08205)(310)$

$\frac{3.815325}{2.50} \rightarrow P = 1.52613$

1.53 atm

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

0.500 mol

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

4

$0.100 / 0.500 =$ **1/5** or **20%**

(c) Calculate the partial pressure (P_{O_2}) of oxygen gas in the mixture. *(3 pts)*

$1.25 \text{ atm} \times 0.20 =$ **0.25 atm**

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

THE MOLE FRACTIONS WILL STAY THE SAME, AS NITROGEN GAS WILL CONTINUE TO INCREASE AT THE SAME 4:1 RATIO AS OXYGEN GAS.

7.5

Chemistry 3A – Fall 2025 Tuesday Version: **Section 25810**

Name Galilea Leon ID 09094104

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

a $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
e $PV = nRT$
b "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
C $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

(b) What is standard temperature in Celsius (°C)? (1 pt)

(c) Convert that standard temperature into Kelvin (K). (1 pt)

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

$PV = nRT$ $P = \frac{nR}{PV}$

$T =$

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

2 In KMT, the temperature and kinetic energy are related. The hotter it gets, the faster the gas particles collide.

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

4.5

$PV = nRT$

$P = \frac{nRT}{V}$

moles ✓

$P = \frac{(0.150)(0.08205)(310)}{2.50}$

$P = 1.52613$

≈ 1.526

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles ($\boldsymbol{n}_{tot}$) of gas in the container? *(1 pt)*

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

(c) Calculate the partial pressure ($\boldsymbol{P}_{O_2}$) of oxygen gas in the mixture. *(3 pts)*

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

Chemistry 3A – Fall 2025 -4 Tuesday Version: **Section 25810**

Name Israe Mack ID [redacted] 0936354

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

C $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
A $PV = nRT$
E "Equal volumes of all gases, at the same temperature and pressure, contain the same number of molecules."
B $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

ATM

(b) What is standard temperature in Celsius (°C)? (1 pt)

(c) Convert that standard temperature into Kelvin (K). (1 pt)

3. Charles's Law Calculation (5 pts)

A flexible balloon is filled to a **volume** of **3.50 L** at room **temperature** (**25°C**). It is placed in a warm environment until its **volume** expands to **4.25 L** at constant pressure. What is the **new temperature** of the balloon in **Celsius** (°C)? (Are intermediate conversions necessary!?)

2

$\frac{3.50L}{25°C} = \frac{4.25L}{x}$ → $(4.25)\frac{3.50L}{25°C} = x$

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

The gas particles will "wiggle" transferring energy and bouncing off each other

5. Ideal Gas Law Calculation (5 pts)
Calculate the pressure (in atmospheres) exerted by 0.150 moles of carbon dioxide gas (CO_2) confined in a 2.50 L container at a temperature of 310 K. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.400}$ moles of nitrogen gas (N_2) and $\mathbf{0.100}$ moles of oxygen gas (O_2). The total pressure of the gas mixture is measured to be $\mathbf{1.25}$ atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

0.500 mol total

(b) Determine the mole fraction (χ) of oxygen gas (O_2). *(2 pts)*

4/1

(c) Calculate the partial pressure (P_{O_2}) of oxygen gas in the mixture. *(3 pts)*

(d) If the total pressure inside the container increases while keeping the relative amounts of each gas constant, what happens to the mole fraction of O_2? Briefly explain why. *(2 pts)*

16.5

Chemistry 3A – Fall 2025 Thursday Version: **Section 25811**

Name Nhia Lee ID 08842411

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

D	$V = k \times n$
E	$PV = nRT$
C	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
A	$P_1V_1 = P_2V_2$

4

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 / 22.4 °C

(b) What is standard temperature in Celsius (°C)? (1 pt)

22.4 °C

(c) Convert that standard temperature into Kelvin (K). (1 pt)

22.4 + 273 = 295.4 K

0.5

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C**. The gas is heated at constant pressure until its new volume reaches **3.60 L**. What is the *new* **temperature** of the gas in **Celsius** (°C)? (Are intermediate conversions necessary?)

5

$T_2 = V_2 \frac{T_1}{V_1}$

$3.60 \times \frac{(22 + 273)}{2.80\ L}$

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$3.60\ L \times \frac{295\ K}{2.80\ L} = 379$

379 − 273 = 106 °C

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

2 they bounce off of each other

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by **0.150 moles** of carbon dioxide gas (CO_2) confined in a **2.50 L** container at a **temperature** of **310 K**. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

5

$\frac{PV}{V} = \frac{nRT}{V}$

$P = \frac{nRT}{V}$ $\quad \frac{0.150 \text{ moles } CO_2 \times \left(0.08205 \frac{L \cdot atm}{mol \cdot K}\right) \times 310 \text{ K}}{2.50 \text{ L}} = 1.53 \text{ atm}$

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of **0.300** moles of helium gas (He) and **0.200** moles of neon gas (Ne). The total pressure of the gas mixture is measured to be **1.50** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)*

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why *(2 pts)*

19

Chemistry 3A – Fall 2025 Thursday Version: **Section 25811**

Name Abel Figueroa ID 0831812

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

d $V = k \times n$
e $PV = nRT$
C $\frac{P_1}{T_1} = \frac{P_2}{T_2}$
a $P_1V_1 = P_2V_2$

4

(a) Boyle's Law $P_1V_1 = P_2V_2$
(b) Charles' Law $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
(c) Gay-Lussac's Law $\frac{P_1}{T_1} = \frac{P_2}{T_1}$
(d) Avogadro's Law
(e) Ideal Gas Law $PV = nRT$

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 STP 1 atm

3

(b) What is standard temperature in Celsius (°C)? (1 pt)

0°C

(c) Convert that standard temperature into Kelvin (K). (1 pt)

273.15 °k

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C.** The gas is heated at constant pressure until its new volume reaches **3.60 L.** What is the *new* **temperature** of the gas in **Celsius (°C)**? (Are intermediate conversions necessary?)

4

V_1 2.80 L

T_1 22°C ≈ 295.15

V_2 3.60 L

$T_1 \cdot T_2 \frac{V_1}{T_1} = \frac{V_2}{T_2} \cdot T_1 T_2$ $T_2 = \frac{V_2 T_1}{V_1}$

$\frac{(2.80\,L)(295.15)}{3.60\,L} = 229.56\,k$

$229.56\,k - 273.15 = \boxed{-43.59\,°C}$

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

1.5 The energy creates pressure

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by **0.150 moles** of carbon dioxide gas (CO_2) confined in a **2.50 L** container at a **temperature** of **310 K**. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$\frac{PV}{V} = nRT$ $P = \frac{nRT}{V}$ $P = \frac{(0.150 mol)(0.08205 \frac{L \cdot atm}{mol \cdot K})(310 K)}{2.50 L}$

$= 1.52 atm$

4.5

6. Partial Pressures and Mole Fraction (8 pts) $P_{total} = P_1 + P_2 + P_3$
A gaseous mixture in a sealed container consists of **0.300** moles of helium gas (He) and **0.200** moles of neon gas (Ne). The total pressure of the gas mixture is measured to be **1.50** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

1.50 mol ≈ 1.50 atm ✓

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

✓

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)*

1.50 mol − 0.300 = 1.2 mol ✓

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why *(2 pts)*

2 ✓

It would decrease since Helium would fill the container more

PV=nRT

Chemistry 3A – Fall 2025 23.5 Thursday Version: **Section 25811**

Name Robert D Ruiz ID 6075374

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

d	$V = k \times n$
e	$PV = nRT$
c	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
a	$P_1V_1 = P_2V_2$

4

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 atm

(b) What is standard temperature in Celsius (°C)? (1 pt)

standard temp is ~~Kelvin~~ C = K + 273.15 ? standard temp is 70°C?

1.5

(c) Convert that standard temperature into Kelvin (K). (1 pt)

70°C + 273.15 = 343.15 K

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C**. The gas is heated at constant pressure until its new volume reaches **3.60 L**. What is the *new* **temperature** of the gas in **Celsius** (°C)? (Are intermediate conversions necessary?)

22°C + 273.15 = 295.15 K

Charles Law $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$\frac{2.80\,L}{295.15\,K} = \frac{3.60\,L}{X}$

$X \cdot \frac{2.80\,L}{295.15\,K} = 3.60\,L$

$X = \frac{3.60\,L \cdot 295.15\,K}{2.80\,L}$

$X = 1.28571 \cdot 295.15\,K$

$X = 379.478\,K$

-273.15

$T_2 = 106°C$

5

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

3 The energy of a gas particle remains constant during collisions. The particle continues to move about its container.

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by **0.150 moles** of carbon dioxide gas (CO_2) confined in a **2.50 L** container at a **temperature** of **310 K**. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$PV = nRT$
$P = \frac{nRT}{V}$

$P = \frac{0.150\ mol \cdot 0.08205 \frac{L \cdot atm}{mol \cdot K} \cdot 310\ K}{2.50\ L} = 46.5 \cdot 0.08205\ K \cdot atm \cdot \frac{1}{2.50}$

$= 46.5 \cdot 0.08205\ atm \cdot \frac{1}{2.50} = 1.52613\ atm$

3 significant figures

5 **1.53 atm**

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of **0.300** moles of helium gas (He) and **0.200** moles of neon gas (Ne). The total pressure of the gas mixture is measured to be **1.50** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

1 $n_{tot} = 0.300\ mol\ He + 0.200\ mol\ Ne = 0.500$

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

2 $\chi_{Ne} = \frac{2}{5} = \frac{0.200}{0.200 + 0.300}$

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)*

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why *(2 pts)*

2 The mol fraction ~~wou~~ of neon would decrease due to the concentration of helium being increased.

Chemistry 3A – Fall 2025 25 Thursday Version: **Section 25811**

Name Samantha Gonzalez ID 09078994

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

D $V = k \times n$	**(a)** Boyle's Law $V_1/T_1 = V_2/T_2$
E $PV = nRT$	**(b)** Charles' Law
C $\frac{P_1}{T_1} = \frac{P_2}{T_2}$	**(c)** Gay-Lussac's Law
A $P_1V_1 = P_2V_2$	**(d)** Avogadro's Law
	(e) Ideal Gas Law

4

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 mol/22.4L

1

(b) What is standard temperature in Celsius (°C)? (1 pt)

0°C

1

(c) Convert that standard temperature into Kelvin (K). (1 pt)

273°K

2

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C**. The gas is heated at constant pressure until its new volume reaches **3.60 L**. What is the *new* **temperature** of the gas in **Celsius** (°C)? (Are intermediate conversions necessary?) No intermediate conversions necessary because final temp is in °C

3

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \quad T_2 \cdot \frac{2.80L}{22°C} = \frac{3.60L}{T_2} \cdot T_2$$

$$T_2 = 3.60L / (2.8L/22°C) = 28.2857°C$$

$$\boxed{T_2 = 28.3°C}$$

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

3 The energy of gas particles during their collisions with each other & the container walls is conserved and will have more energy when more temperature is added that excites them.

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by **0.150 moles** of carbon dioxide gas (CO_2) confined in a **2.50 L** container at a **temperature** of **310 K**. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$\frac{PV}{V} = \frac{nRT}{V}$ $P = \frac{nRT}{V}$

$P = \frac{0.150 mol\ CO_2 \cdot 0.08205 \frac{L \cdot atm}{mol \cdot K} \cdot 310 K}{2.50 L}$

5

$P = 1.52613 atm$

$\boxed{P = 1.53 atm}$

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of **0.300** moles of helium gas (He) and **0.200** moles of neon gas (Ne). The total pressure of the gas mixture is measured to be **1.50** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

$0.300 moles + 0.200 moles = \boxed{0.500 moles}$

8

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

$\frac{0.200 mol\ Ne}{0.500 mol_{tot.}} = \boxed{0.400 = \chi_{Ne}}$

$PV = NRT$

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)*

$\frac{P_{Ne}}{P_{total}} = \frac{n_{Ne}}{n_{tot}} = \frac{0.200 mol}{0.500 mol} = 0.400$

$1.50 atm \cdot .400 = 0.1600$

$\boxed{P_{Ne} = 0.1600 atm}$

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why *(2 pts)*

The mole fraction of neon will decrease because there is a higher total number of moles in the container.

Chemistry 3A – Fall 2025 18 Thursday Version: **Section 25811**

Name Vio Tarlva ID 1081907

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

d $V = k \times n$	
e $PV = nRT$	
B $\frac{P_1}{T_1} = \frac{P_2}{T_2}$	
C $P_1V_1 = P_2V_2$	

2

(a) Boyle's Law
~~**(b)**~~ Charles' Law
~~**(c)**~~ Gay-Lussac's Law
~~**(d)**~~ Avogadro's Law
~~**(e)**~~ Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

R, 0.082058 $\frac{L \cdot atm}{(mol \cdot K)}$

(b) What is standard temperature in Celsius (°C)? (1 pt)

K − 273.15 K = C

K = C + 273.15K
−273.15K

(c) Convert that standard temperature into Kelvin (K). (1 pt)

K = C + 273.15 K

1

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C**. The gas is heated at constant pressure until its new volume reaches **3.60 L**. What is the *new* **temperature** of the gas in **Celsius** (°C)? (Are intermediate conversions necessary?)

22°C + 273.15K = 295.15 K

$V_1 = 2.80L$
$T_1 = 22°C = 295.15K$
$V_2 = 3.60L$
$T_2 = ?$

4.5

$T_2 = \frac{2.80L \cdot 3.60L}{295.15K}$

$P_2 \cdot \frac{P_1}{T_1} = \frac{P_2}{T_2} \cdot P_2 \cdot T_2$

$T_2 = \frac{295K \cdot 3.60L}{2.80L} = 379K$

$T_2 = \frac{T_1 \cdot P_2}{P_1}$

penguin :o

4. Kinetic Molecular Theory (3 pts)

According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

3

When energy of gas particles collide with each other, they do not change each other/affect each other because they are undergoing a process called elasticity. =o→o=

5. Ideal Gas Law Calculation (5 pts)

Calculate the **pressure** (in atmospheres) exerted by **0.150 moles** of carbon dioxide gas (CO_2) confined in a **2.50 L** container at a **temperature** of **310 K**. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$PV = nRT$

$$\frac{P(2.50\,L)}{2.50\,L} = \frac{(0.150\,mol\,CO_2)(0.08205\,\frac{L \cdot atm}{mol \cdot K})(310\,K)}{(2.50)L}$$

all units cancel except for atm :o

Find: P
given: .150 mol (n) CO_2
given: 2.50 L, V
given: 310 K
given: R

4.5

$P = 1.5261$ atm

6. Partial Pressures and Mole Fraction (8 pts)

A gaseous mixture in a sealed container consists of **0.300** moles of helium gas (He) and **0.200** moles of neon gas (Ne). The total pressure of the gas mixture is measured to be **1.50** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

given: 0.300 mol He
given: 0.200 mol Ne
given: 1.50 atm mix

$0.300_{mol\ He} + 0.200$ mol Ne $= 0.500$ mol

1

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

$n = n_1 + n_2$

idk sorry! :/

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)*

sorry

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why *(2 pts)*

2

The mol fraction of neon will be less concentrated in the container since a surplus of Helium has been added to the container. It does not mean the moles were destroyed, ever, just that Helium has a greater concentration now. idk if that's right, But I tried anyways. :/

14.5

Chemistry 3A – Fall 2025 Thursday Version: **Section 25811**

Name Fernanda Dominguez ID 09811154

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

d $V = k \times n$	**(a)** Boyle's Law
e $PV = nRT$	**(b)** Charles' Law
c $\frac{P_1}{T_1} = \frac{P_2}{T_2}$	**(c)** Gay-Lussac's Law
a $P_1V_1 = P_2V_2$	**(d)** Avogadro's Law
	(e) Ideal Gas Law

4

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

22.7 ?

(b) What is standard temperature in Celsius (°C)? (1 pt)

The standard temperature in Celsius is C° = K - 273

1

(c) Convert that standard temperature into Kelvin (K). (1 pt)

K = C + 273

The standard temperature converted to Kelvin is ____

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C**. The gas is heated at constant pressure until its new volume reaches **3.60 L**. What is the *new* **temperature** of the gas in **Celsius** (°C)? (Are intermediate conversions necessary?)

5

$V_1 = 2.80L$

$T_1 = 22°C + 273 = 295$

$V_2 = 3.60L$

$T_2 = ?$

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$

$\frac{2.80L}{295} = \frac{3.60L}{T}$

$3.60 \cdot 0.009491525 = 3.60L \cdot T_2$

$3.60 \quad 105.357429 = 3.60L$

$379.285 = T_2$

-273

$106°C = T_2$?

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

1 what happens to the energy of gas particles during the collision with eachother is sep / N/A

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by **0.150 moles** of carbon dioxide gas (CO_2) confined in a **2.50 L** container at a **temperature** of **310 K**. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$PV = nRT$

2.5 $P(2.50) = (0.150 mol)(0.08205 \frac{L \cdot atm}{mol \cdot K})(310 K)$

$\frac{2.50}{2.50} = \frac{3.815325}{2.50}$

$P = 9.5382125$

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of **0.300** moles of helium gas (He) and **0.200** moles of neon gas (Ne). The total pressure of the gas mixture is measured to be **1.50** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

1 The total number of moles (n_{tot}) of gas is 0.500 ? N/A?

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

$6.0221 \times 10^{21}((0.200) = 0.200$? N/A?

1×0.200

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)*

$P = 1.50 atm$

$Ne = 0.200 mol$ N/A?

$n = 300 mol$

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why *(2 pts)*

N/A ?

19.5

Chemistry 3A – Fall 2025 Thursday Version**: Section 25811**

Name Maria Gomez ID ?

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

D	$V = k \times n$ (n: moles)
E	$PV = nRT$
C	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
A	$P_1V_1 = P_2V_2$

4

(a) Boyle's Law $P_1V_1 = P_2 \cdot V_2$
(b) Charles' Law $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
(c) Gay-Lussac's Law $\frac{P_1}{T_1} = \frac{P_2}{T_2}$
(d) Avogadro's Law $\frac{V_1}{n_1} = \frac{V_2}{n_2}$
(e) Ideal Gas Law $PV = nRT$

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

22.4 atm

(b) What is standard temperature in Celsius (°C)? (1 pt)

0°C

2

(c) Convert that standard temperature into Kelvin (K). (1 pt)

Kelvin conversion °C + 273.15

0 + 273.15 = 273.15

3. Charles's Law Calculation (5 pts)

$\frac{V_1}{T_1} = \frac{V_2}{T_2}$ V_2

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C**. The gas is heated at constant pressure until its new volume reaches **3.60 L**. What is the *new* **temperature** (T_2) of the gas in **Celsius (°C)**? (Are intermediate conversions necessary?)

22°C + 273.15 = 295.15

3

Known:
$V_1 = 2.80\,L$
$T_1 = 22°C$
$V_2 = 3.60\,L$

Unknown
T_2

$T_2 = \frac{V_1 \cdot V_2}{T_1}$

$T_2 = \frac{2.80\,L \cdot 3.60\,L}{22\,(C)}$ $\left[\frac{2.80\,L \cdot 3.60\,L}{295} = \frac{10.08}{295} = 0.034169\right]$

/

0.45818

= 0.458

KMT, remains same?

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.
Energy of the gas particles is lost during their collisions

1

5. Ideal Gas Law Calculation (5 pts) P-pressure V-volume n-moles R-constant T-temp $PV = nRT$
Calculate the **pressure** (in atmospheres) exerted by **0.150 moles** of carbon dioxide gas (CO_2) confined in a **2.50 L** container at a **temperature of 310 K**. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$PV = nRT$
$P = ?$
$V = 2.50L$
$n = 0.150 moles$
$R = 0.08205$
$T = 310K$

4.5

$P = \frac{(0.150 moles)(0.08205)(310K)}{2.50L}$ $\frac{3.815325}{2.50L} = 1.526130$ 1.53

$P = 1.526 atm$

3.815

6. Partial Pressures and Mole Fraction (8 pts) Xgas
A gaseous mixture in a sealed container consists of **0.300** moles of helium gas (He) and **0.200** moles of neon gas (Ne). The total pressure of the gas mixture is measured to be **1.50** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

$P_{total} = P_1 + P_2 + P_3 \ldots$

0.300 moles He + 0.200 moles Ne = 0.5 moles

1

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)* Ne / total

$\frac{0.200 moles\ Ne}{0.5 moles\ total} = 0.4$

?

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)* $P = X_{gas} \times total$?

0.4 x X gas = ?

daltons law partial pressure.

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why *(2 pts)*

If more helium gas is injected in at constant temp & volume, The mole fraction of neon would decrease.

18.5

Chemistry 3A – Fall 2025 Thursday Version: **Section 25811**

PV = nRT all Laws $V_2 = \frac{V_1 \cdot T_2}{T_1}$

Name Z Martinez ID 1004784

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

Boyles $V_2 = \frac{V_1 T_1}{T_2}$

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

d	$V = k \times n$
e	$PV = nRT$
b	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
c	$P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

$\frac{PV}{nT} = R$ 2

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

1 atm which equals 760 torr or 760 mmHg or 100,325 Pa

(b) What is standard temperature in Celsius (°C)? (1 pt)

1 75°C ? or is it -275.15

(c) Convert that standard temperature into Kelvin (K). (1 pt)

348.15 K 0 ?

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C**. The gas is heated at constant pressure until its new volume reaches **3.60 L**. What is the *new* **temperature** of the gas in **Celsius** (°C)? (Are intermediate conversions necessary?)

2.5

$T_2 = \frac{V_1 \times T_1}{V_2}$

22°C + 273.15 = 295 K

$T_2 = \frac{2.80\,L \times 295\,K}{3.60\,L} = 229.44\ K$

229 K ⇒ −44 °C

229.44 − 273.15 = −43.71 °C

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

They don't lose any kinetic energy.

3

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by **0.150 moles** of carbon dioxide gas (CO_2) confined in a **2.50 L** container at a **temperature** of **310 K**. (Use $R = 0.08205 \frac{L \cdot atm}{mol \cdot K}$)

$PV = nRT$

$P(2.50\ L) = (0.150\ mol)\left(0.08205 \frac{L \cdot atm}{mol \cdot K}\right)(310\ K)$

$P(2.50\ L) = (0.150)(0.08205\ atm)(210)$

$\frac{P(2.50\ L)}{2.50} = \frac{2.584575\ L \cdot atm}{2.50\ L} = 1.03383\ atm$ or $1.03\ atm$

3.5

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of **0.300** moles of helium gas (He) and **0.200** moles of neon gas (Ne). The total pressure of the gas mixture is measured to be **1.50** atm.

$P_{total} = 1.50\ atm$

$n_1 = 0.300$ He, 0.200 Ne

$n_2 = 0.500\ mol$

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

0.500 moles

1

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

$n\chi = \frac{0.200}{0.500} = 0.4$?

2

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)*

$P_{Ne} = \frac{P_{total} \times n_1}{n_2}$ $\frac{1.50\ atm \times 0.300\ mol}{0.500\ mol} = \frac{0.45\ atm}{0.500} = 0.9\ atm$

1.5

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why *(2 pts)*

It decreases due to the addition He taking up more of the mole fraction, since the constant temp & volume allow for no unaccounted changes in pressure.

2

10

Chemistry 3A – Fall 2025 Thursday Version: **Section 25811**

Name Cynthia Sanchez ID 1093672

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

D $V = k \times n$	**(a)** Boyle's Law
E $PV = nRT$	**(b)** Charles' Law
C $\frac{P_1}{T_1} = \frac{P_2}{T_2}$	**(c)** Gay-Lussac's Law
A $P_1V_1 = P_2V_2$	**(d)** Avogadro's Law
	(e) Ideal Gas Law

4

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

.0815

(b) What is standard temperature in Celsius (°C)? (1 pt)

32

(c) Convert that standard temperature into Kelvin (K). (1 pt)

0.5

32 + 273.15 = 305.15 K

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C**. The gas is heated at constant pressure until its new volume reaches **3.60 L**. What is the *new* **temperature** of the gas in **Celsius** (°C)? (Are intermediate conversions necessary?)

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

2.5 Pressure increases, it is elastic (kinectic energy) no energy is lost.

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by **0.150 moles** of carbon dioxide gas (CO_2) confined in a **2.50 L** container at a **temperature** of **310 K**. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

2.5 $.08205 \frac{2.5 \cdot .atm}{.150 \times 310} =$ atm

4.56 = atm

$.08205 \frac{2.5}{4.5} \times 100 =$

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of **0.300** moles of helium gas (He) and **0.200** moles of neon gas (Ne). The total pressure of the gas mixture is measured to be **1.50** atm.

(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

0.5 .06

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

.06/1.50

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)*

$\times \frac{.300}{200}$ P = .04

.6/1.50 =

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why *(2 pts)*

0.5

Chemistry 3A – Fall 2025 Thursday Version: **Section 25811**

Name Canna Mendoza ID 07187628

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

A $V = k \times n$	~~(a)~~ Boyle's Law
B $PV = nRT$	**(b)** Charles' Law
E $\frac{P_1}{T_1} = \frac{P_2}{T_2}$	**(c)** Gay-Lussac's Law
D $P_1V_1 = P_2V_2$	~~(d)~~ Avogadro's Law
	(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

(b) What is standard temperature in Celsius (°C)? (1 pt)

293 - 1

0.5

(c) Convert that standard temperature into Kelvin (K). (1 pt)

273 + 1

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C.** The gas is heated at constant pressure until its new volume reaches **3.60 L.** What is the *new* **temperature** of the gas in **Celsius** (°C)? (Are intermediate conversions necessary?)

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by $\mathbf{0.150}$ **moles** of carbon dioxide gas (CO_2) confined in a $\mathbf{2.50}$ **L** container at a **temperature** of $\mathbf{310}$ **K**. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.300}$ moles of helium gas (He) and $\mathbf{0.200}$ moles of neon gas (Ne). The total pressure of the gas mixture is measured to be $\mathbf{1.50}$ atm.
(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)*

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why
(2 pts)

I didn't study :(1.5

Chemistry 3A – Fall 2025 Thursday Version: **Section 25811**

Name Nadia Diaz ID 09896716

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

a	$V = k \times n$
d	$PV = nRT$
e	$\frac{P_1}{T_1} = \frac{P_2}{T_2}$
C	$P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

P

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

(b) What is standard temperature in Celsius (°C)? (1 pt)

F-32)/1.8 ?

(c) Convert that standard temperature into Kelvin (K). (1 pt)

0.5 °C+273.15 ?

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C**. The gas is heated at constant pressure until its new volume reaches **3.60 L**. What is the *new* **temperature** of the gas in **Celsius** (°C)? (Are intermediate conversions necessary?)

1 2.80/22 x 3.60 = 0.458181818 ?

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by $\mathbf{0.150}$ **moles** of carbon dioxide gas (CO_2) confined in a $\mathbf{2.50}$ **L** container at a **temperature** of $\mathbf{310}$ **K**. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.300}$ moles of helium gas (He) and $\mathbf{0.200}$ moles of neon gas (Ne). The total pressure of the gas mixture is measured to be $\mathbf{1.50}$ atm.
(a) What is the total number of moles (n_{tot}) of gas in the container? *(1 pt)*

(b) Determine the mole fraction (χ) of neon gas (Ne). *(2 pts)*

(c) Calculate the partial pressure (P_{Ne}) of neon gas in the mixture. *(3 pts)*

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why
(2 pts)

Chemistry 3A – Fall 2025 Thursday Version: **Section 25811**

Name Alyssa Garza ID 09901566

Mini-Midterm on Gases and Gas Law **Total Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

____ $V = k \times n$
____ $PV = nRT$
____ $\frac{P_1}{T_1} = \frac{P_2}{T_2}$
____ $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

(b) What is standard temperature in Celsius (°C)? (1 pt)

(c) Convert that standard temperature into Kelvin (K). (1 pt)

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of **2.80 L** at a **temperature** of **22°C**. The gas is heated at constant pressure until its new volume reaches **3.60 L**. What is the *new* **temperature** of the gas in **Celsius** (°C)? (Are intermediate conversions necessary?)

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by $\mathbf{0.150}$ **moles** of carbon dioxide gas (CO_2) confined in a $\mathbf{2.50}$ **L** container at a **temperature** of $\mathbf{310}$ **K**. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.300}$ moles of helium gas (He) and $\mathbf{0.200}$ moles of neon gas (Ne). The total pressure of the gas mixture is measured to be $\mathbf{1.50}$ atm.
(a) What is the total number of moles ($\boldsymbol{n}_{\text{tot}}$) of gas in the container? *(1 pt)*

(b) Determine the mole fraction ($\boldsymbol{\chi}$) of neon gas (Ne). *(2 pts)*

(c) Calculate the partial pressure ($\boldsymbol{P}_{\text{Ne}}$) of neon gas in the mixture. *(3 pts)*

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why
(2 pts)